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Deputy Directorate General of Quality Control

Plasma Smelting Laboratory-3-H

TEST REPORT

Test Report No: TR (023)/PS/22

Job Card No: JC (044)/PS/22

Date: 27-04-22

INDENTER'S DATA	
Name: Fraz-ur-Rehman	Designation: Manager
Group & Site: MP/3C	Directorate: ADP
Organization: NDC	Contact No: 3627

GENERAL DATA		
Sample Name: Silica Phenolic	ID No: PP-10A-Abl-01 (Sample 01)	
Lot No: Si (820)-10A/1120-221~333	Lot Size: 8000 RM	
Quantity: 01	Chemical Composition: Composite	
Test Type: General	Sample Receiving Date: 26-04-22	Test Date: 26-04-22

TEST DATA	
Test Name: Ablation Test	
Qualification Standard: NDC-Abl/QC/Si-C Ablatives/02 (Rcv#1)	
Standard Test Method: 1-E 458-72(Re approved 2002)	
Test Environmental Conditions: Ambient	
Test/Calorimetry Equipment & Instruments Used: Huels Arc Heater, Digital Micrometer.	
Radiation & Optical Equipment used: Nil	Enthalpy Measurement Technique: By Empirical Formula
Flow Temperature:	3062 K
Flow Enthalpy:	3.603 MJ/Kg
Duration of Exposure:	20 Sec
Results:	▪ Test results attached (at page 02 of 02)
Remarks:	▪ These results relates only to the sample mentioned and tested above. ▪ This test report shall not be reproduced except in full, without written approval of the laboratory. ▪ The working medium for generation of Plasma Arc is air. ▪ Uncertainty in ablation rate calculation is ± 0.00774 mm/Sec at 95 % confidence level.

Test Performed By: Wasim Ahmed Zahid, GM (Elect)	27-4-22
Reviewed By: Muhammad Zahid Irfan, PD (PS)	27-4-22
Contact No: 3205/3287	

Approved By: Khalid Atta, Dir (STE)	28-4-22
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Prepared By: FM/GM	Reviewed & Recommended By: GM/HOD	Approved By: GMR/HOD/PD
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Test Results

Sr #	Description	Measured Value	Qualification Criteria Standard Value	Status
1	Average initial thickness before test	11.064 mm	NA	Qualified
2	Final residual thickness after test	11.020 mm	NA	
3	Maximum ablation in terms of thickness	0.044 mm	NA	
4	Ablation Rate in terms of thickness	0.002 mm/Sec	≤ 0.15 mm / Sec	
5	Max Back Surface Temperature	NA	NA	

Additional Remarks: Nil

Test Performed By:

Wasim Ahmed Zahid, GM (Elect)

End of Report

Prepared By: FM/GM	Reviewed & Recommended By: GM/HOD	Approved By: GMR/HOD/PD
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